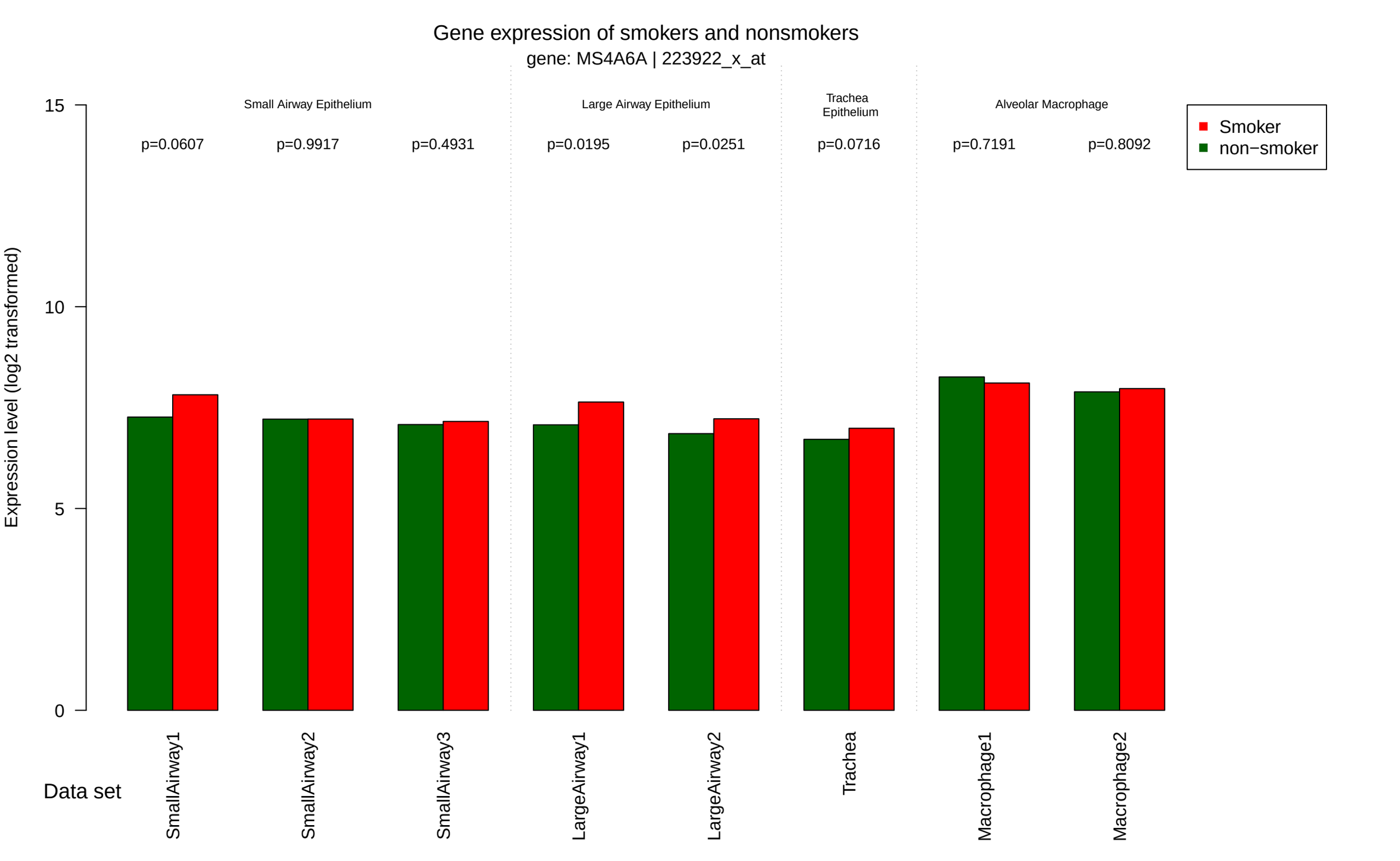
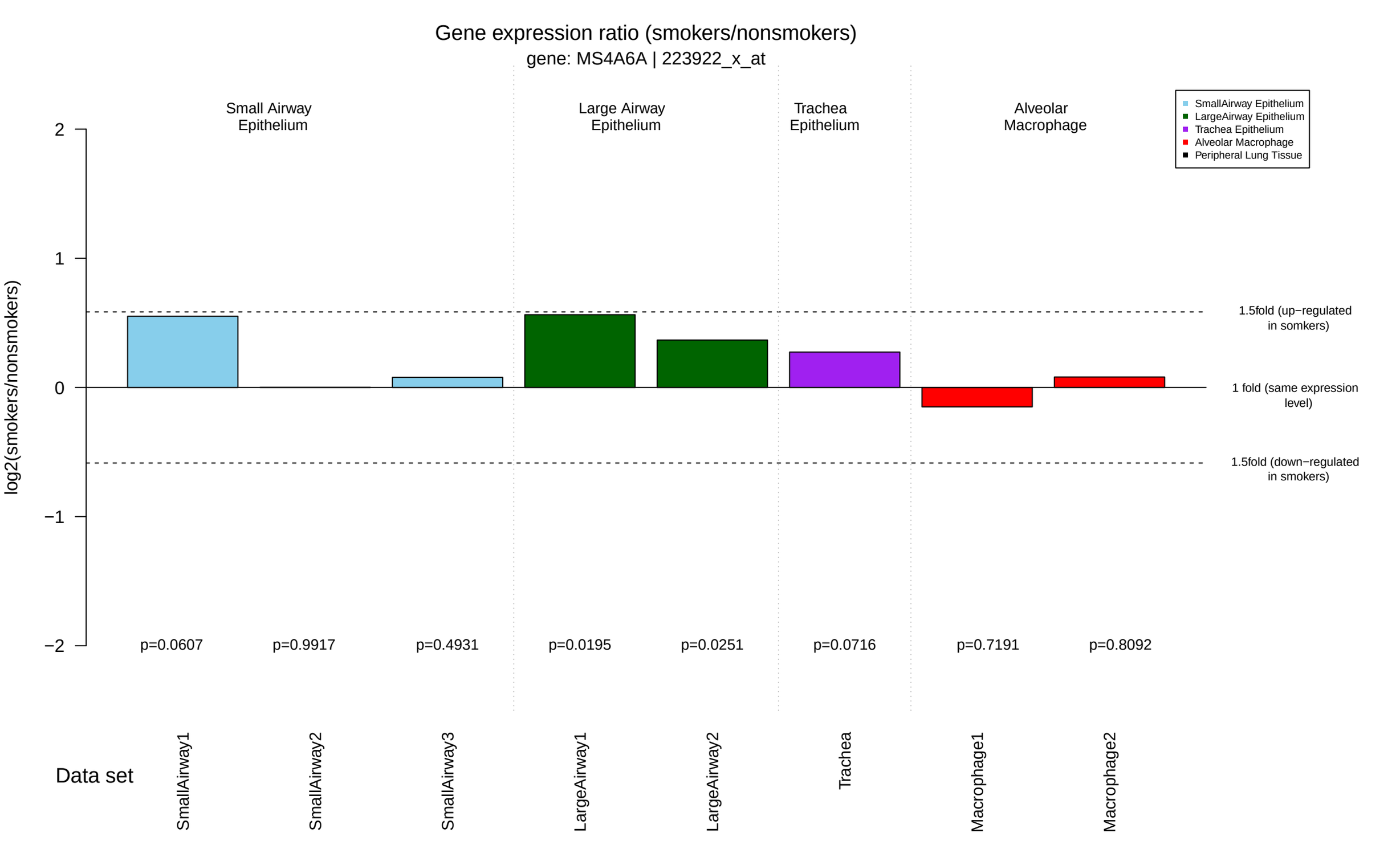
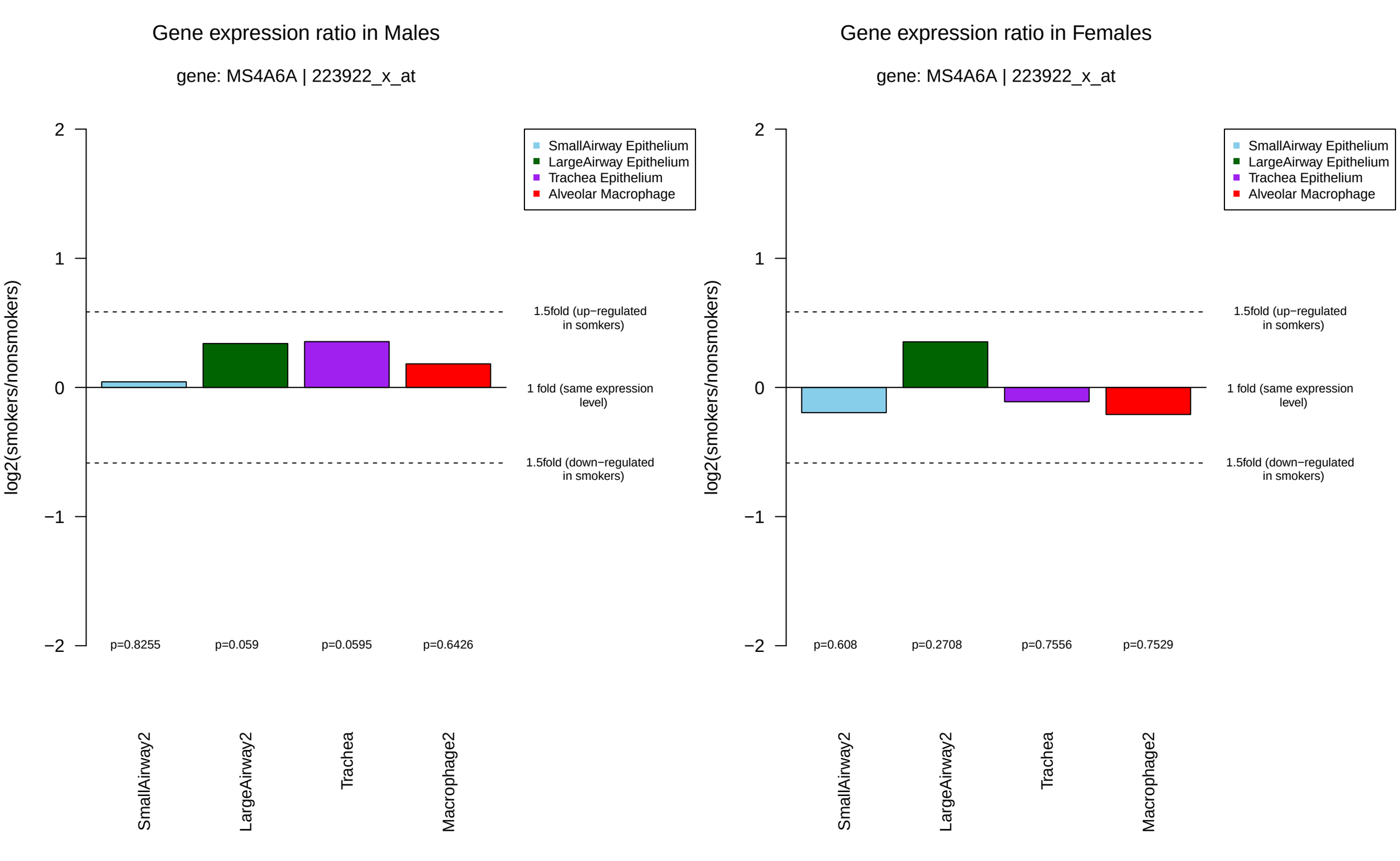
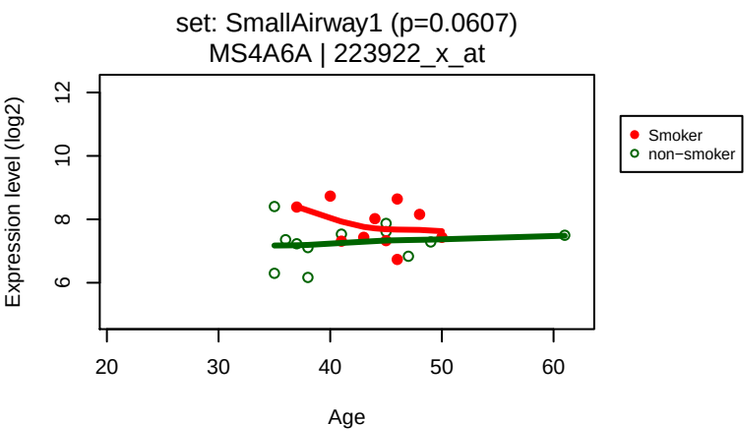
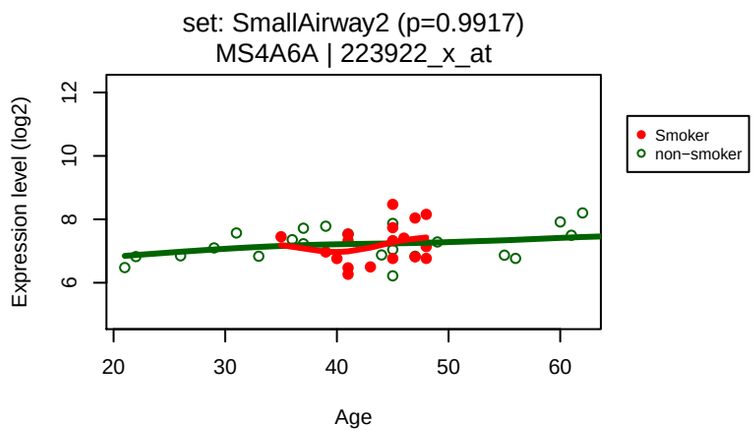
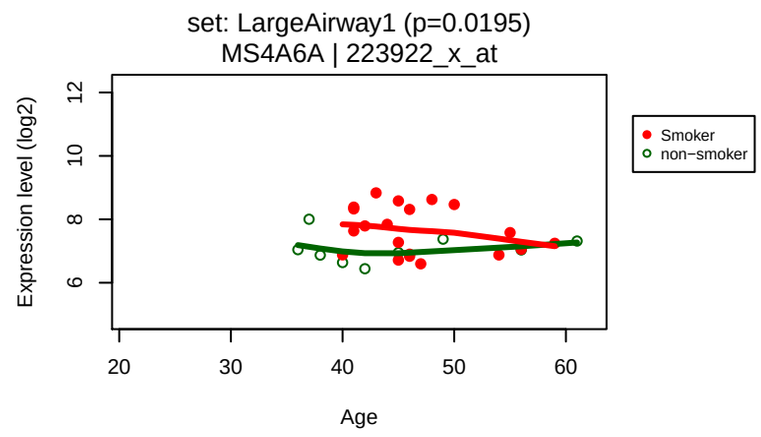

Gene expression of smokers and nonsmokers
gene: MS4A6A | 223922_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0607
p=0.9917
p=0.4931
p=0.0195
p=0.0251
p=0.0716
p=0.7191
p=0.8092
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

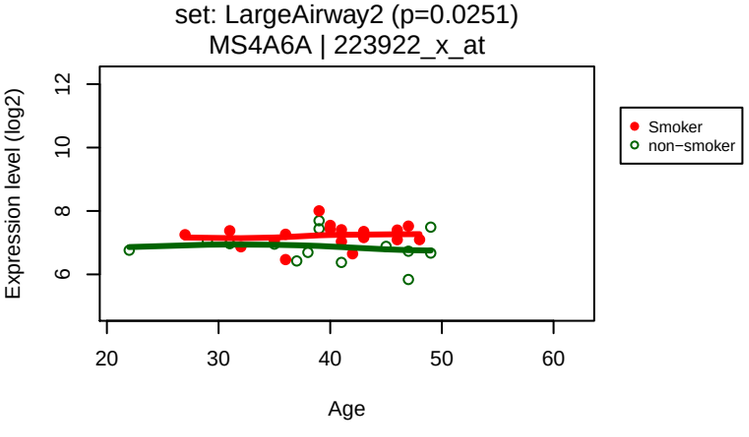

Gene expression ratio (smokers/nonsmokers)
gene: MS4A6A | 223922_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0607
p=0.9917
p=0.4931
p=0.0195
p=0.0251
p=0.0716
p=0.7191
p=0.8092
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

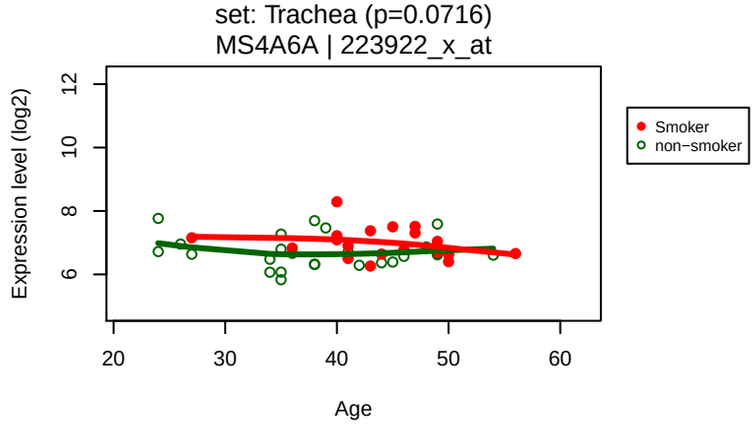

Gene expression ratio in Males
gene: MS4A6A | 223922_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8255
p=0.059
p=0.0595
p=0.6426
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A6A | 223922_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.608
p=0.2708
p=0.7556
p=0.7529
SmallAirway2
LargeAirway2
Trachea
Macrophage2

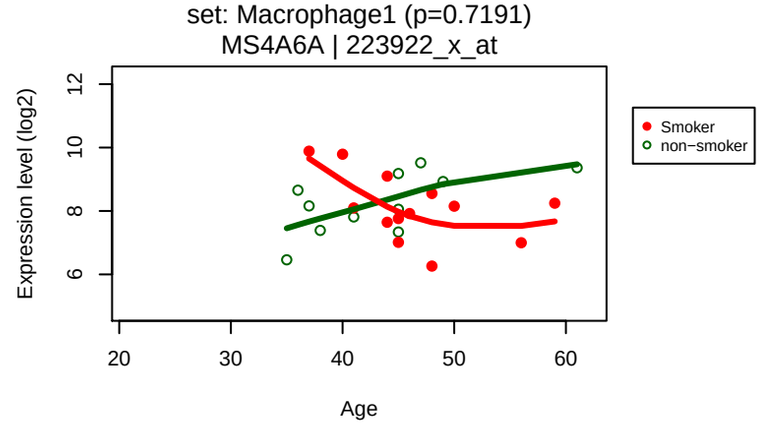

set: SmallAirway1 (p=0.0607)
MS4A6A | 223922_x_at
Expression level (log2)
Age
Smoker
non−smoker

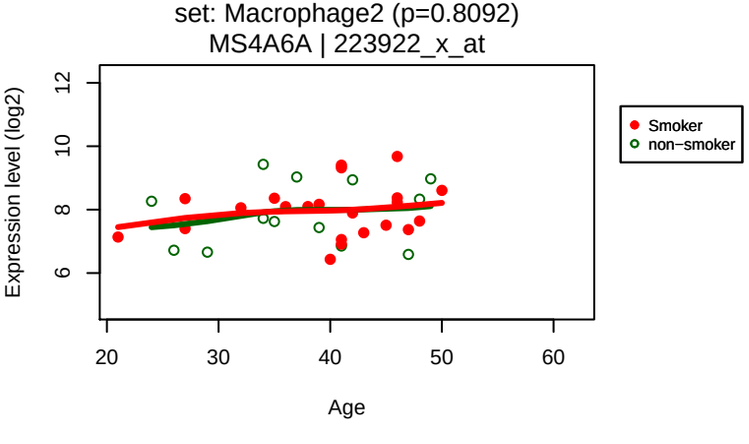

set: SmallAirway2 (p=0.9917)
MS4A6A | 223922_x_at
Expression level (log2)
Age
Smoker
non−smoker

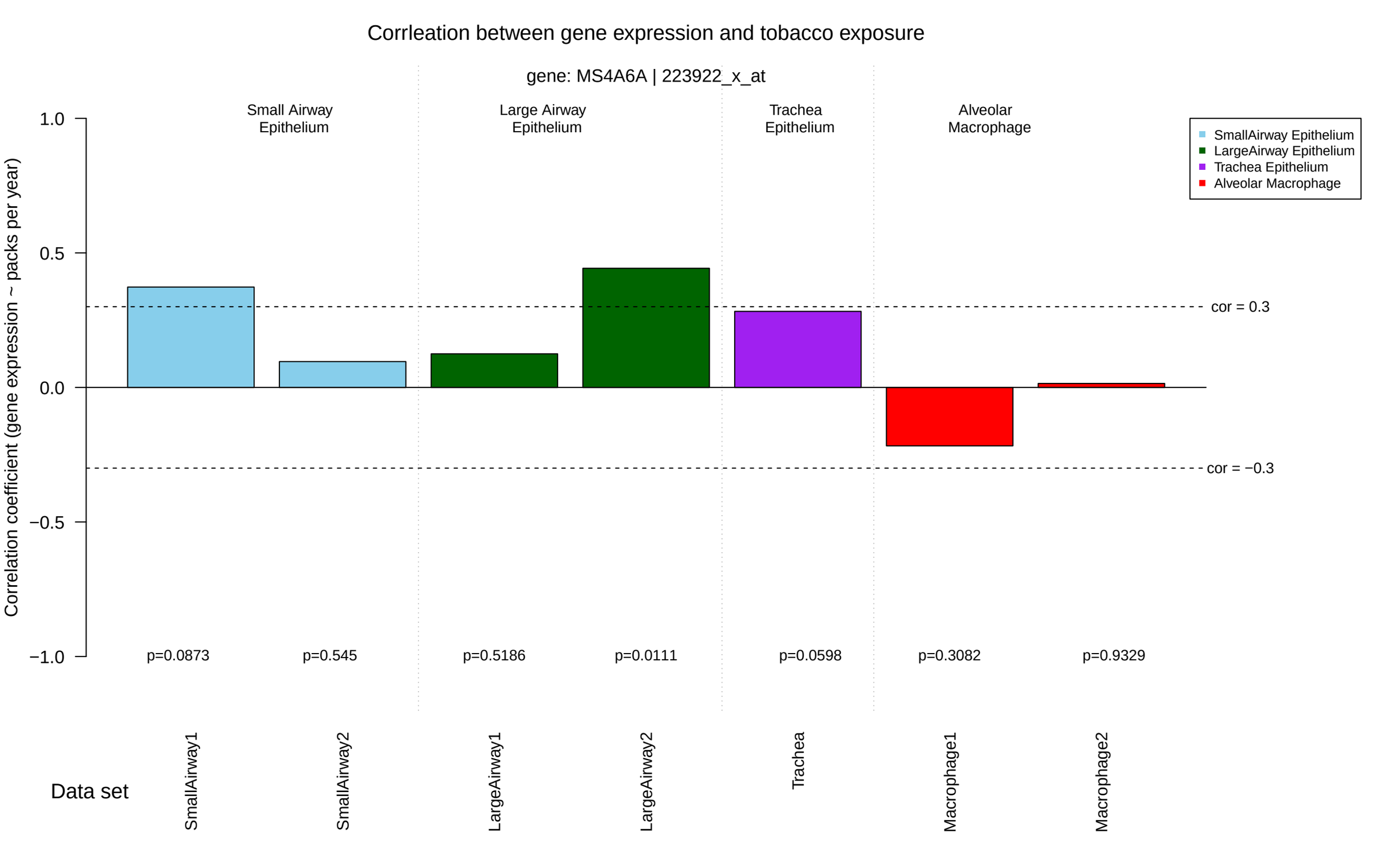

set: LargeAirway1 (p=0.0195)
MS4A6A | 223922_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0251)
MS4A6A | 223922_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0716)
MS4A6A | 223922_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7191)
MS4A6A | 223922_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8092)
MS4A6A | 223922_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A6A | 223922_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0873
p=0.545
p=0.5186
p=0.0111
p=0.0598
p=0.3082
p=0.9329
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2